Evaluating Cultural Competence in a Fine-Tuned Llama 3.3 70B Model for HBCU-Related Educational Context

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Abstract

The use of large language models (LLMs) in educational applications continues to grow; however, conventional evaluations often emphasize task performance and language quality without adequately examining cultural and contextual relevance. This study evaluates cultural competence in HBCU-related educational contexts using Llama 3.3-70B-Instruct fine-tuned using 253 reviewed education-focused question-response pairs. A set of 20 HBCU-related factual queries were answered by both the base and fine-tuned models under matched generation settings. Both response sets were independently rated by Llama 3.2-1B and Gemini 2.5 Flash using the same five-dimensional, five-point REACH cultural-competence rubric. The Llama evaluator's five-criterion mean increased from 2.10 ($SD = 1.02$) for the base model to 2.40 ($SD = 0.94$) for the fine-tuned model, whereas Gemini's mean decreased slightly from 4.87 ($SD = 0.33$) to 4.75 ($SD = 0.83$). Response-level agreement between evaluators was weak for the base responses ($r = .26$) and approximately zero for the fine-tuned responses ($r = -.04$). These mixed results indicate that fine-tuning effects depend strongly on detailed prompting and automated cultural-competence scores should be manually interpreted alongside separate factual verification.

CCS Concepts

• **Computing methodologies** → **Natural language generation**; • **Applied computing** → *Education*; • **Human-centered computing** → *Human computer interaction (HCI)*.

Keywords

Large Language Models, HBCU, Cultural Competence, Llama 3.3 70B, QLoRA, Fine-Tuning, LLM Evaluation, AI in Education

1 Introduction

In recent years, generative artificial intelligence (GenAI) has increasingly been incorporated into education for question answering, learning support, information access, student advising, and instructional assistance [1]. UNESCO highlights that the educational use of GenAI should be human-centered, impartial, accessible, and ethically evaluated rather than assessed merely according to technological functionality [1]. Therefore, educational LLM evaluation should consider not only whether a response is proficient or useful, but also whether it correctly represents the cultural and institutional context of the user.

This concern is relevant to HBCU-related educational contexts. HBCUs are colleges or universities that were established before 1964 with the main purpose of educating Black Americans, but these institutions differ notably in institutional characteristics, programs, locations, resources, and group of students [4]. Therefore, an LLM should not treat HBCUs or their students as homogeneous. Culturally proper responses should protect preexisting ideas, avoid biased perception and unsupported general observation, and identify when an HBCU-specific viewpoint is or is not relevant.

Recent studies also show that cultural awareness remains a challenge for LLMs [2, 3]. Research on cultural awareness identifies a few important components of inclusive language model development such as culturally factual data, model adaptation, and culturally sensitive evaluation [2]. CulturalBench also demonstrates that LLMs can struggle with culturally nuanced questions, supporting the need for explicit cultural evaluation rather than treating general language performance as evidence of cultural competence [3].

Fine-tuning provides one mechanism for adapting a pretrained LLM to specialized contexts. Parameter-efficient methods such as Low-Rank Adaptation (LoRA) reduce the computational cost of domain adaptation by training low-rank parameter updates while retaining the pretrained model weights [5]. QLoRA further reduces memory requirements by combining low-rank adaptation with quantized pretrained models [6]. These methods make domain adaptation feasible even for large models, but successful fine-tuning does not automatically establish cultural competence or factual reliability.

The main contribution of this study is the evaluation of a 70-billion-parameter LLM in an HBCU-related educational setting. The study evaluates responses across five cultural-competence dimensions using two automated evaluators and examines how fine-tuning changes the apparent effect of domain adaptation. Because the evaluation questions are fact-focused, the study also motivates separate source-based factual verification rather than treating cultural-competence scores as evidence of factual accuracy.

2 Literature Review

Cultural awareness in LLMs involves more than retrieving culture-specific factual knowledge. It includes sensitivity to variation within communities, contextual norms, culturally situated communication, and risks associated with stereotyping or oversimplification [2]. Pawar et al. argue that culturally responsive LLM development requires attention to datasets, adaptation methods, benchmarks, and human interaction [2]. CulturalBench similarly evaluates cultural knowledge using human-written and human-verified questions across diverse regions and demonstrates the importance of explicit cultural benchmarking [3].

These concerns are especially important in education, where AI-generated responses may influence information seeking, learning, and decision support. UNESCO therefore recommends a human-centered approach that considers inclusion, equity, cultural diversity, and ethical validation when GenAI is introduced into education [1].

Model scale is another relevant design factor. Meta released Llama 3.3-70B-Instruct as a text-only, instruction-tuned 70-billion-parameter model [7]. Larger capacity may improve knowledge coverage and response quality, but it substantially increases computational requirements. LoRA and QLoRA provide practical mechanisms for adapting such models without full-parameter fine-tuning [5, 6].

Evaluation remains a separate methodological challenge. LLMs are increasingly used as automated judges because they can scale the assessment of open-ended responses, but prior research identifies potential position, verbosity, self-preference, and reasoning biases in LLM-based evaluation [8]. Therefore, conclusions about cultural competence should not depend unquestioningly on a single automated evaluator.

Recent evaluation research further supports assessing LLMs across multiple dimensions rather than relying on a single aggregate measure. HELM emphasizes standardized, multi-metric evaluation across diverse scenarios [9]. Cultural benchmarking extends this principle to culturally situated behavior: CDEval evaluates multiple cultural dimensions across domains [10], Naous et al. identify systematic cultural bias and context-misalignment in multilingual models [11], and Tao et al. show that LLM outputs can reflect cultural value biases while targeted prompting can improve alignment in some settings [12]. Together, these studies motivate treating cultural competence as a distinct, multidimensional property that requires explicit evaluation.

3 Methodology

3.1 Experimental Setup, Data, and Tools

Experiments were conducted on a Google Cloud VM equipped with an NVIDIA A100-SXM4-80GB GPU and approximately 167 GiB of system memory. A 375 GB NVMe working volume was used for large model files and training artifacts. The trainable model was the 4-bit Llama 3.3-70B-Instruct checkpoint loaded with Unsloth and Hugging Face Transformers. QLoRA fine-tuning used LoRA rank 16 and alpha 16, a per-device batch size of 1, gradient accumulation of 8 (effective batch size 8), maximum sequence length 1024, one epoch, and a learning rate of 1×10^{-4}. The software stack included Python, Unsloth, Transformers, TRL, PEFT, bitsandbytes, and related data-processing libraries.

3.2 Overall Project Workflow

Figure 1 summarizes the study workflow from data preparation and fine-tuning through the 20-query matched response generation and two-model automated evaluation.

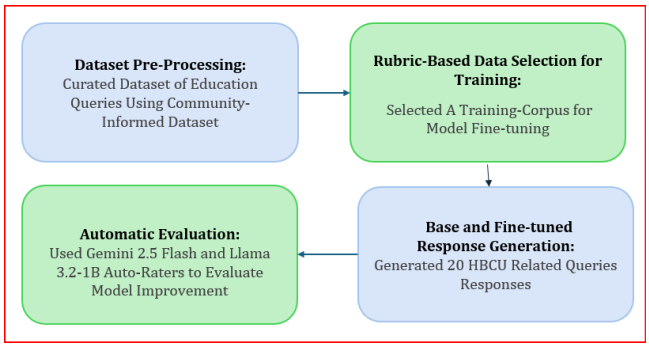


Figure 1: Overall workflow for 253-example dataset preparation, Llama 3.3 70B QLoRA fine-tuning, 20-query matched response generation, and automated cultural-competence evaluation.

3.2.1 Dataset Pre-Processing. The Google Reach workshop dataset contained approximately 1,832 question-response pairs covering multiple subject areas. To create a dataset appropriate for fine-tuning an education-focused model, the data was filtered to identify questions directly or indirectly related to education. Relevant topics included teaching, learning, student advising, academic programs, educational technology, student engagement, HBCU student experiences, campus culture, career preparation, student services, and academic inclusion.

A combination of Python-based keyword filtering and manual review was used to identify questions relevant to education. After initially filtering queries labeled as educational, a manual review was conducted to identify questions that were indirectly related to education and may have been overlooked by keyword-based filtering. This process also helped remove false positives. Relying solely on the original topic labels proved insufficient, as some relevant educational queries were categorized under other topics, which limited the number of suitable queries available for training.

Following the selection process, the dataset was manually cleaned to improve its overall quality and consistency. Duplicate and nearly identical queries were removed, along with unrelated questions and incomplete or irrelevant responses. Formatting inconsistencies, such as unnecessary spaces, quotation marks, and line breaks, were also standardized to create a cleaner and more consistent dataset for fine-tuning.

Although manual and automated filtering improved data quality, the process was subject to potential classification errors and reviewer subjectivity. Additionally, cleaning for relevance and formatting did not guarantee the factual accuracy of every response; therefore, separate fact-checking and evaluation procedures were conducted during later stages of the study.

3.2.2 Rubric-Based Data Selection for Training. The REACH Cultural Competence Rubric was used to assess the cultural competency of model-generated responses. Each response was rated across five criteria on a 1–5 scale, ranging from poor (1) to excellent (5). Higher scores represent responses that demonstrate greater cultural awareness, contextual accuracy, respect, and linguistic appropriateness. The five criteria measure the following:

- **Framing and Narrative Integrity:** Evaluates whether a response accurately represents community perspectives and experiences while considering historical context, power imbalances, and potential biases in dominant narratives.
- **Nuanced and Intersectional Representation:** Measures whether a response recognizes diversity within cultural groups and considers how intersecting factors, such as race, gender, socioeconomic status, disability, age, or geography, can shape individual experiences.
- **Respect of Cultural Boundaries:** Assesses whether a response appropriately handles culturally sensitive, sacred, private, or protected knowledge while respecting a community's cultural and intellectual boundaries.
- **Task and Situational Appropriateness:** Evaluates whether a response is relevant to the specific question and appropriately incorporates cultural, local, and situational context without making unsupported assumptions or presenting implausible information.
- **Language Comprehension and Alignment:** Measures whether the response uses respectful, contemporary, and contextually appropriate language while avoiding outdated terminology, forced imitation of community language, or language that is unsuitable for the situation.

The five-dimensional cultural-competence rubric was used during evaluation to screen candidate examples for cultural appropriateness, contextual relevance, and response quality. The final training corpus contained 253 reviewed education-focused question-response pairs. These examples were used to adapt Llama 3.3-70B-Instruct with QLoRA. The training prompt instructed the model to answer questions directly and accurately, to apply HBCU-specific context when relevant, and to avoid inventing institutional, historical, statistical, or location-specific details when uncertain.

3.2.3 Base and Fine-Tuned Response Generation. An evaluation set of 20 new HBCU-related factual queries was used after fine-tuning. The questions covered HBCU identification, locations, institutional history, founders, and counts or lists of HBCUs. The test questions were kept separate from the 253-example training corpus, and the training workflow included exact and near-duplicate checks to reduce train-test leakage. Each question was presented to both the base and fine-tuned Llama 3.3-70B-Instruct models using the same system prompt and deterministic generation settings. Because this held-out set emphasized direct institutional and historical facts, it provided limited coverage of nuanced and intersectional HBCU student experiences.

3.2.4 Automated Evaluation. Both the 20 base responses and the 20 fine-tuned responses were evaluated using the same five-dimensional REACH rubric on a 1–5 scale. Llama 3.2-1B served as a local automated evaluator, and Gemini 2.5 Flash served as an independent external evaluator through Google Cloud. Each evaluator received the same question-response pairs and the same scoring definitions. Ratings were summarized using criterion-level means and standard deviations, five-criterion response means, and response-level Pearson correlations between the two evaluators. Automated ratings were interpreted cautiously because LLM-as-a-judge systems can exhibit model-specific biases [8]. Cultural-competence scores were treated as distinct from factual accuracy; a response-level source-based fact-checking layer is therefore needed to pair each cultural-competence judgment with an independent factual-accuracy assessment.

Table 1: Comparison of Base and Fine-Tuned Model Ratings Across Cultural-Competence Criteria ($N = 20$)

Criterion	Base Llama M (SD)	Fine-tuned Llama M (SD)	Base Gemini M (SD)	Fine-tuned Gemini M (SD)
Framing and Narrative Integrity	1.05 (0.22)	1.05 (0.22)	4.85 (0.37)	4.75 (0.91)
Nuanced Representation	1.90 (1.07)	2.05 (1.00)	4.85 (0.37)	4.75 (0.91)
Cultural Boundaries	2.20 (1.28)	2.90 (1.45)	4.95 (0.22)	4.80 (0.89)
Task Appropriateness	2.30 (1.38)	2.50 (1.19)	4.75 (0.64)	4.60 (1.10)
Language Alignment	3.05 (2.01)	3.50 (1.91)	4.95 (0.22)	4.85 (0.49)
Five-criterion mean	**2.10 (1.02)**	**2.40 (0.94)**	**4.87 (0.33)**	**4.75 (0.83)**

4 Results and Discussion

The two automated evaluators produced different conclusions about the effect of fine-tuning. The Llama 3.2-1B autorater assigned a higher five-criterion mean to the fine-tuned model ($M = 2.40$, $SD = 0.94$) than to the base model ($M = 2.10$, $SD = 1.02$), a mean increase of 0.30 points. In contrast, Gemini 2.5 Flash assigned a slightly lower mean to the fine-tuned model ($M = 4.75$, $SD = 0.83$) than to the base model ($M = 4.87$, $SD = 0.33$), a decrease of 0.12 points. Thus, the new 20-query experiment does not support a single evaluator-independent conclusion that fine-tuning improved cultural competence.

At the criterion level, the Llama evaluator showed its largest fine-tuning increases for Respect of Cultural Boundaries (+0.70) and Language Comprehension and Alignment (+0.45), followed by Task Appropriateness (+0.20) and Nuanced Representation (+0.15); Framing and Narrative Integrity was unchanged. Gemini showed small decreases for all five criteria, ranging from -0.10 to -0.15. Gemini also produced substantially higher absolute scores than Llama in both conditions, indicating a strong evaluator-specific scoring effect.

Evaluator agreement was limited. The response-level Pearson correlation between Llama and Gemini overall scores was $r = .26$ for the base responses and $r = -.04$ for the fine-tuned responses; across all 40 rated responses, $r = .03$. These near-zero associations indicate that the two evaluators did not rank response quality consistently.

Gemini ratings were also concentrated near the upper end of the scale, suggesting a possible ceiling effect, whereas Llama produced a wider range of lower scores.

Due to the 20 evaluation questions emphasize culturally-appropriate responses ratings alone are insufficient to determine whether fine-tuning reduced hallucinations and factual HBCU information. A response can be culturally respectful and well aligned while still containing an incorrect date, institution, founder, location, or count. Accordingly, the revised evaluation should report a separate response-level factual-accuracy judgment for each base and fine-tuned response using authoritative institutional or government sources, alongside the REACH cultural-competence scores.

Qualitative examples further illustrate why aggregate means should be supplemented with response-level inspection. In a stronger fine-tuned response to the query: "What HBCUs are in Maryland?", the model correctly listed Bowie State University, Coppin State University, Morgan State University, and the University of Maryland Eastern Shore and provided concise institutional context; Gemini rated this response 5.0 overall, while Llama rated it 3.0. In contrast, the fine-tuned response to the query: "What HBCUs are located in Washington, D.C.?", conflated Howard University with several non-HBCU institutions and even included entries explicitly marked as errors; Gemini rated that response 1.4 overall, while Llama rated it 3.0. These examples reinforce both the need for factual verification and the observed disagreement between automated evaluators.

5 Limitations

The evaluation used 20 generated HBCU-related questions, so the findings should be treated as a pilot study rather than a complete assessment of HBCU cultural competence and factual reliability. The 253 question-response training set is also relatively small compared with the diversity of HBCU histories, institutions, and educational contexts. In addition, the held-out questions focused mainly on direct factual knowledge and did not adequately test intersectional experiences involving factors such as gender, socioeconomic background, first-generation status, disability, or geography. Future evaluation should therefore include targeted intersectional prompts rather than relying primarily on institutional fact questions. The two automated evaluators also produced different scoring patterns, and the study did not include human cultural-competence raters with direct HBCU experience. This is an important limitation: automated ratings alone cannot fully validate cultural appropriateness. Future work should include HBCU-experienced human raters, assess inter-rater agreement, and compare human judgments with Llama and Gemini scores while reporting factual accuracy separately.

6 Conclusion

Fine-tuning Llama 3.3 70B on 253 reviewed education-focused examples produced mixed results across the two automated evaluators. Llama 3.2-1B rated the fine-tuned responses higher overall than the matched base responses (2.40 vs. 2.10), whereas Gemini 2.5 Flash rated them slightly lower (4.75 vs. 4.87). The largest positive changes under the Llama evaluation occurred for cultural boundaries and language alignment, while Gemini showed small declines across all five criteria. The weak evaluator agreement (base $r = .26$; fine-tuned $r = -.04$) demonstrates that conclusions about culturally responsive model behavior depend strongly on the evaluator. The 70B experiment therefore shifts the paper's emphasis away from model size alone and toward the joint importance of domain adaptation, matched evaluation, factual verification, and evaluator reliability. A stronger evaluation framework should combine automated cultural-competence scores, response-level factual-accuracy judgments, qualitative examples, intersectional HBCU prompts, and human ratings from individuals with relevant HBCU experience before claiming that fine-tuning improves HBCU-related educational responses.

GenAI Usage Disclosure

GenAI tools were used in this work for research, writing, and programming support. OpenAI ChatGPT was used to assist with language editing, manuscript organization, LaTeX formatting, and research-related guidance. Anthropic Claude was used to assist with the development, review, and debugging of code used in the research workflow. Llama 3.2-1B and Gemini 2.5 Flash were used as automated evaluators, as described in the Methodology section. All AI-assisted text, code, analyses, and outputs were reviewed and verified by the authors, who take full responsibility for the final manuscript and research results.

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